

riff



Type a vibe. Get **real, editable** music.

@zeromusicbot make a funky disco loop, 120bpm



a ZeroClaw music director · zeromusicbot#0169 · github.com/gitayam/riff-music-bot

THE PROBLEM

AI music is a black box.

You get an audio file you can't read, can't change, and can't learn from. No score, no knobs, no *why*. Want it darker, or a bassline added? Roll the dice again and hope.

WHAT IT DOES

Talk to it. Get a track **and** its code.

Every reply is three things: ① the Strudel code, ② a 🎙️ voice message of the track, and ③ one line on the choices. Then shape it by talking — "darker", "add a bassline", "give me 3 variations".

```
setcpm(78/4)
stack(
  sound("bd ~ sd ~").bank("RolandTR909"),
  sound("hh*8").gain(.4),
  note("a2 c3 e3 g3").sound("sawtooth").lpf(800),
  n("0 2 4 6").scale("A:minor").sound("piano").room(.4)
) // real Strudel — loops in the browser, every line editable
```

HOW IT WORKS

Local, end to end.

No cloud render. The model writes the pattern; a parse-gate refuses anything that won't play; the **real strudel.cc engine** renders it to audio headlessly; it ships as a Discord voice message.



Other teams can call it too – `POST /generate {"prompt"} → {code, audio}` over HTTP.

THE MUSIC

A loop → sections → a song.

One **loop** is the unit. Label loops as **intro · verse · chorus · outro**, then merge and layer them — energy comes from **adding parts and opening the filter**, not just volume. Here's that arc *measured from a real render*:

SECTION	MEAN VOL	HIGHS (HATS / LEAD)
intro (0–8s)	–32.9 dB	just chords
verse (8–24s)	–18.9 dB	drums in
chorus (24–40s)	–14.1 dB	peak — hook + full kit
outro (40–48s)	–32.3 dB	winds down

The chorus lands **~19 dB louder and far brighter** — the textbook shape. And when a track wants words, Riff speaks a line over the beat with **OpenAI TTS**.

Tight loops. Verify everything.

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- | | | |
|----|--------|---|
| 01 | soul | Gave the agent a music-director persona + a cited theory cheat-sheet, so it composes like a producer. |
| 02 | gate | The model wrote code that wouldn't play → a parse-gate that rejects bad patterns before they ship. |
| 03 | model | Rate-limits took it down → switched to gpt-5.4 with a fallback chain (→ mini → mistral). |
| 04 | render | Headless Strudel → real audio, auto-delivered as voice messages by a decoupled watcher. |
| 05 | ship | Public repo, one-command setup, a 16/16 health check, and a demo runbook. |
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Hardest bug: "it didn't work" → root cause wasn't the code, it was an **unstarted process**. The fix made the failure *impossible to miss* — the health check now flags it.

BUILT TO NOT BREAK

Resilient by design.

Fallback chain

One model hiccup never kills it — gpt-5.4 → gpt-5.4-mini → Mistral.

Won't ship junk

The parse-gate guarantees every posted track actually plays.

Always on

Three launchd services: the bot, the voice-message watcher, the HTTP API.

One command

`strudel-doctor.sh` verifies the whole stack — 16/16 before any demo.

Secrets stay out

Keys live only in a gitignored `.env` — scanned before every push.

Self-hostable

Clone, `setup.sh`, run. No cloud render dependency.

TRY IT

Real music. Real code. In chat or over HTTP.

bot [zeromusicbot#0169](#) — @mention it in Discord

repo github.com/gitayam/riff-music-bot

try “make a chill lofi loop” → code + a 🎤 voice message

⌚ Temporary demo — the bot runs on a laptop for the hackathon, so it may be offline. The code is fully self-hostable.

Build. Ship. Shortcuts.